

Genomic architecture of ancestry sorting: diagnostic index, not recombination, governs its direction

Draft results summary (Module A, architecture). Regenerated at the clean pipeline conventions (pooled-parental MAF gate 0.15, sorting threshold $\tau = 0.6$, near-fixation floor $\phi = 0.85$, cM05 clustering; diagnostic index ungated), reproduced by `moduleA.sorting/R/moduleA.architecture.R`. Methods are deferred to Supplementary Materials. All findings are descriptive: any recombination effect awaits the recombination-matched neutral null, because a non-recombining autosomal block does not have lower N_e than its surroundings under neutrality.

Results

Diagnostic index is separable from recombination and reflects reduced diversity. Across the genome the diagnostic index (DI) is essentially orthogonal to the map-interpolated recombination rate (Spearman $\rho = -0.03$) and to pooled parental minor-allele frequency ($\rho = -0.02$), making it a genuinely independent predictor of sorting. Its dominant correlate is *low* within-species diversity ($\rho = -0.46$ with π), not elevated absolute divergence ($\rho = +0.13$ with d_{xy}), so DI is largely a reduced-diversity differentiation signal rather than a divergence-island one. A classical differentiation-island signature—jointly elevated F_{ST} and d_{xy} , reduced π , and very large linkage blocks—is confined to the lowest-recombination decile (Fig. 1a), yet DI is flat across recombination and does not track it. DI and recombination are therefore separable axes.

Sorting is depleted, not enriched, in low-recombination regions. At the independent-unit level, the fraction of units that sort is *lowest* in the least-recombining decile (0.04 versus ~ 0.37 elsewhere; Fig. 1b), because those regions consist of a few very large linkage blocks whose consensus washes out. The higher SNP-level value in that decile (0.15) relative to the unit level (0.04) is spatial pseudo-replication: adjacent, redundant SNPs in a single low-recombination block. Consistently, sorting magnitude *increases* with recombination (prop.fixed $\sim$ recombination, standardised slope $+0.05$, $z = 112$). Sorting is thus not concentrated where recombination is low—contrary to a naive linked-selection expectation, and consistent with neutral book-keeping (fewer independent units per low-recombination block) pending the null. Along the genome, the same blocks stand out as SNP-level sorting peaks coinciding with elevated local linkage disequilibrium (Supplementary Fig. S4). These large blocks are also internally heterogeneous in diagnostic content: within-cluster DI spread increases with cluster size, from near-zero in marker pairs to a plateau below the genome-wide marker spread in the largest clusters (Fig. 2), so cluster-level diagnostic index is aggregated as the maximum over member markers—tagging a block as diagnostic when any member is.

Direction is governed by diagnostic index; recombination adds only a weak independent pull. In a logistic model of sorting direction among directionally resolved units, $P(\text{aquilonia}) \sim \text{DI} + \text{recombination} + \text{parental MAF} + \log(\text{block size})$, diagnostic index dominates (standardised coefficient $+1.57$, $z = 120$): high-DI units fix toward *F. aquilonia*, low-DI toward *F. polychaeta*. Recombination contributes a small but significant independent effect in the opposite-to-expected-noise direction—low recombination leans *F. aquilonia* after DI, parental MAF and size are controlled (coefficient -0.09 , $z = -6.8$)—a modest linked-selection lead. It is, however, roughly seventeen-fold weaker than DI, and at the raw unit level direction is nearly flat across recombination (Fig. 1c); the stronger low-recombination *F. aquilonia* excess seen at the SNP level is largely pseudo-replication that flattens once independent units are counted. This polarity is robust to the classification

thresholds: both the low-DI-toward-*F. polycytena* / high-DI-toward-*F. aquilonia* reversal along the DI axis (Supplementary Fig. S2) and the dominance of the DI coefficient (Supplementary Fig. S3) persist across the sorting threshold τ and the near-fixation floor ϕ .

Outlook. Module A establishes that the predictable polarity of ancestry sorting is organised by locus diagnosticity, a reduced-diversity axis orthogonal to recombination, and that sorting is depleted rather than enriched in low-recombination regions once pseudo-replication is removed. These conclusions are insensitive to the sorting-classification thresholds (Supplementary Figs. S1–S3). A weak residual low-recombination pull toward *F. aquilonia* survives control for DI—a lead for the intrinsic (linked-selection / incompatibility) hypothesis. Adjudicating it requires the recombination-matched, haplodiploid neutral null; the intrinsic and extrinsic drivers are then discriminated, respectively, by among-population linkage disequilibrium between unlinked diagnostic units and by overlap with climate genotype–environment outliers.

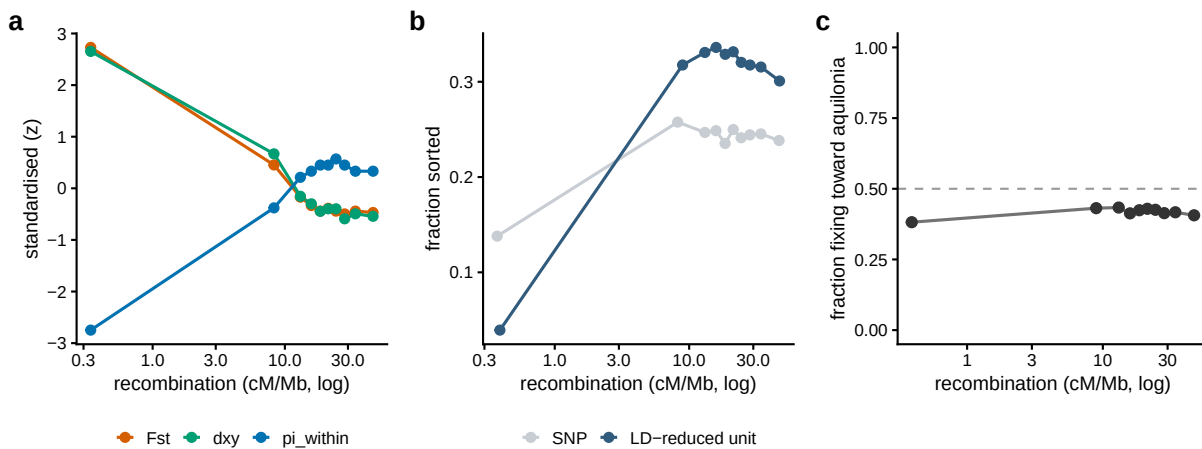


Figure 1: **Genomic architecture of sorting.** (a) Standardised relative differentiation (F_{ST}), absolute divergence (d_{xy}) and within-species diversity (π) across recombination deciles: the differentiation-island signature is confined to the lowest-recombination decile. (b) Fraction of loci sorted versus recombination at the SNP level and at the independent-unit level; the unit-level collapse in low recombination is the removal of spatial pseudo-replication. (c) Direction of directionally resolved sorting (fraction fixing toward *F. aquilonia*) versus recombination at the unit level: the diagnostic-index polarity is not mirrored by a recombination gradient. Regenerated by `moduleA.sorting/R/moduleA.architecture.R`; robustness to the sorting thresholds is shown in Supplementary Figs. S1–S4.

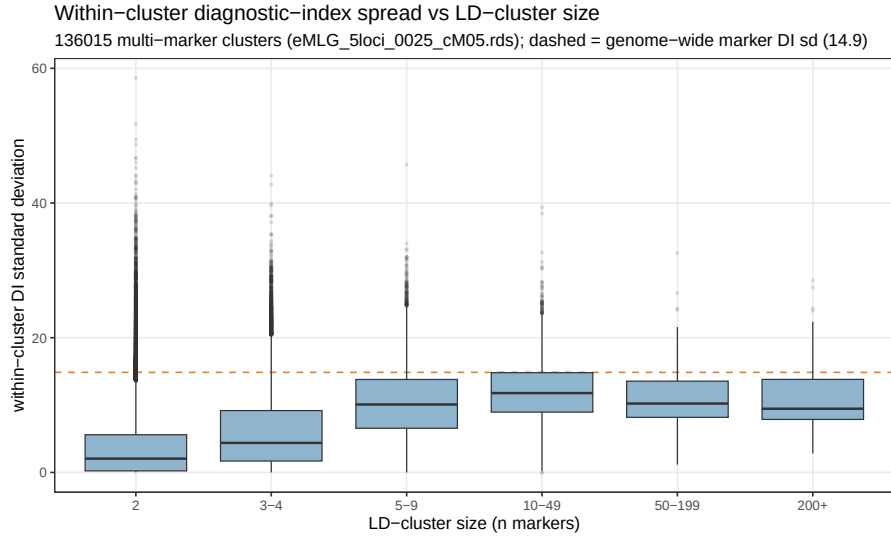


Figure 2: **Within-cluster diagnostic-index spread increases with LD-cluster size.** Standard deviation of the diagnostic index (DI) among the member markers of each multi-marker LD cluster (cM05 clustering; $n = 136,015$ clusters with ≥ 2 markers), by cluster-size class. Marker pairs are nearly DI-homogeneous (median SD ≈ 2); spread rises to ≈ 10 – 12 for clusters of 5–49 markers and plateaus for larger blocks, while the within-cluster DI *range* keeps growing (to a median of 66 for the largest). Every size class stays below the genome-wide marker DI standard deviation (dashed, 14.9), so clusters remain more DI-homogeneous than random markers, but large low-recombination blocks mix diagnostic and non-diagnostic loci (Spearman $\rho = 0.52$ between cluster size and within-cluster DI SD). This motivates summarising cluster diagnostic index as the maximum over member markers.